8 Fundamentals of Experimentation

Abstract

The design and analysis of experiments were developed by R. A. Fischer, a famous biostatistician, during the 1920s and 1930s. Other key authors, including Campbell and Stanley also contributed to understanding the experiments. Experimental studies have become increasingly complex over the years, involving more complex data sets (big data) due to the advanced statistical methods. In this Chapter, we present the fundamentals of experimentation and explain the concepts of experimentation, the experimental process and the concept of experimental error and controlling "noise" in experiments. We follow this with three main principles of experimentation: "the 3Rs," randomization, replication, and local control." The final section of this Chapter discusses 14 elements of designing effective experiments, including, among others, the selection of experimental design and treatments; treatment of response variables; the role of uniformity trials; management of experimental; and experimental protocol.

Keywords

Agronomy experiment · Experimental error · Experimental site · Experimental process · Field experiment · Laboratory experiments · On-station research · Local control or blocking · Randomization · Response variable · Replication · Three Rs · Treatments · Treatment error · Unexplained variation or "Noise" · Uniformity Trials

Learning Outcomes

By the end of this Chapter, the learner is expected to be able to:

1. Appreciate the origin of agricultural experiments.
2. Recognize the common types of agricultural experiments.
3. Illustrate the basic experimental process.
4. Differentiate between "experimental error" and "treatment error".
5. Demonstrate understanding of "Extraneous variables" to determine how to reduce "Noise" in an experiment.
6. Relate the principle of "local control" to data validity.
7. Apply the principle of 'Three R" in designing an effective experiment.
8. Identify the main causes of experimental error in an agronomy experiment.
9. Determine a suitable number of replicates in an experiment.
10. Design an effective experimental protocol.

8.1 Design of Experiments

8.1.1 Brief History of the Design of Experiments

Design and analysis of experiments were developed by Sir R. A. Fischer, a famous biostatistician while working at "Rothamsted Experimental Station," then the Institute of Arable Crops Research in the 1920s and 1930s (Yates 1964; Fisher 1980). Founded in 1943, the experimental station is one of the oldest global agricultural research institutions. Its primary mission was to evaluate the impact of inorganic and organic fertilizers on crop yields. Fischer dedicated 14 years to this station, conducting extensive crop experiments. Through the analysis of its immense data, Fisher laid the foundation for experimentation. He developed the concept of analysis of variance (ANOVA) and explained it as an important analysis for partitioning the variation present in the data into two components: (1) due to attributable factors, and (2) due to chance factors (Fisher 1980).

Fishers' initial contribution primarily focused on applying statistical methods in agriculture, advancing many of the statistical concepts scientists use today. For example, His books advanced the concept of randomly assigning individu-

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als to groups and replication in experiments (Hall 2007). The recognition of the need to systematically address variability in experiments led to the formulation of three foundational principles (3Rs): randomization, replication, and local control (Reduce noise). These principles became the cornerstone of robust experimental design. Other statistical procedures advanced were testing the significance of differences, critical value, and chi-square goodness of fit. These methods shaped the field of experimentation and statistical analysis and have been widely applied in "biological, psychological, and agricultural" research.

Several other publications have made significant contributions to the understanding of research experimental designs. Among them are works by Campbell and Stanley (1963), Cook and Campbell (1979) and Shadish et al. (2002). These authors emphasized the importance of experimental designs in establishing causal relationships between variables and delineated the essential characteristics of true experiments and quasi-experimental designs. They also paid particular attention to validity and how validity can be enhanced, outlining the various types of experimental designs and discussing threats to validity. Their books are considered essential reading for anyone interested in experimental research design. This applies to both social and natural sciences.

Campbell and Stanley (1963) outline potential threats to validity in experimental designs. They specified 15 different types and evaluated all designs regarding "potential threats to validity." They also discuss the importance of rigorous research design and analysis techniques, such as matching techniques and statistical controls, to address these threats. Overall, researchers continue to draw upon the information presented in these classical books to ensure the scientific rigor and credibility of their research.

Experimental studies have become increasingly complex over the past few decades, thanks to the emergence of advanced statistical techniques and computer technology. This progress in computing technology and statistical methodology has played a vital role in managing and analyzing large datasets from intricate experimental designs. In recent years, the concept of big data science has gained popularity, and experimental studies now involve analyzing even more massive and complex datasets. With the aid of advanced statistical methods, researchers can extract meaningful patterns and insights from vast amounts of data.

This Chapter and Chap. 9 discuss the nature of experiments.

8.1.2 Classes of Experiments

There are four overarching classes of experiments in agriculture as shown hereby:

1. *Laboratory experiments*: These are where the independent variable is intentionally manipulated, and the dependent variable is measured within a controlled and artificial setting (Brüggemann and Bizer 2015). This method effectively manages and controls extraneous variables that might influence the dependent variable. They can be a study carried out in a science laboratory or a screen house experiment (greenhouse experiment), where conditions are controlled. In some greenhouses, the conditions, mainly temperatures, soil moisture, and humidity, are controlled and monitored.
2. *Field experiments*: These are studies conducted in a natural environment, where the independent variables are manipulated, and measurements of the dependent variable are taken on test units in their authentic, real-world setting, such as a farm. Depending on the specific research site, field experiments can be further distinguished as "*on-station and on-farm*" experiments. The distinction lies in whether the experiments are conducted within controlled research facilities or directly on farms.
3. *On-station research*: It is a type of research carried out in an environment intentionally designed and controlled by the researcher, often in a research station, university farm or a comparable setting where researchers maintain complete control. While farmers are not actively engaged in managing the research, they might be invited to participate and learn from the process, depending on the study's objectives.
4. *On-farm research*: On-farm research (OFR) entails conducting research directly on farmers' fields. This approach can take various forms: it may be researcher-designed and farmer-managed, entirely designed and managed by researchers or involve collaborative efforts between farmers and researchers. Central components of OFR encompass the farmer, their land, active farmer involvement and the specific environmental conditions prevailing on the farmer's land. On-farm research and experimentation are discussed in Chap. 10.

8.2 Common Types of Laboratory and Field Experiments

Agronomy Experiments Broadly, agronomy experiments involve crop production and soil management. The main aim is to develop management practices that optimize crop production. Experiments mainly include field crop management, soil and water management, cropping systems, eco-physiology, and management of biotic and abiotic stresses. The following are common agronomy experiments:

- *Varietal screening experiments*: Aim to test the performance of different varieties or genotypes. It often involves developing and evaluating new varieties with improved traits, such as disease resistance, drought tolerance, and yields. Depending on the purpose of the screening, various types are recognized. For instance, in maize varietal experiments, screening may involve progeny (progeny trials), observation or preliminary trials, national, regional, and international performance trials, on-farm trials or demonstrations (Badu-Apraku et al. 2012).
- *Forage production experiments*: Forage production is essential to livestock production. These experiments are conducted to develop efficient and sustainable forage production systems. They mainly involve evaluating and comparing different forage varieties, management practices, and environmental conditions to determine the best strategies for maximizing yields and quality of forages.
- *Intercropping experiments*: These experiments focus on assessing the effect of different intercropping and rotation systems on soil health, pest management, and overall yields. They help increase understanding of species interactions and design systems that enhance resource utilization (Fig. 8.1). Conducting these experiments over several years to evaluate the system's stability is advisable. Stability considerations account for variable climatic conditions and pressures from pests and diseases, leading to more resilient agricultural systems (Finney 1990).

Crop Rotation Experiments Experiments primarily investigate opportunities to enhance agricultural production in the short and long term by manipulating the design of crop rotation. Treatments are implemented over multiple cropping seasons or years, and their effects are assessed in the final season (Payne 2015).

Soil Fertility Experiments The discipline of soil science explores the diverse role of soils and investigates ways to use the soil more sustainably. Soil fertility experiments will, therefore, aim to evaluate the influence of soil management practices on the availability of soil nutrients and agricultural

Fig. 8.1 Soil fertility and agronomy experiments; (**a**) soil fertility, (**b**) intercropping of sorghum and green grams in a farmers field

productivity. The experiments provide valuable information on the best management practices to optimize soil fertility and improve crop production by providing nutrients. Typical experiments assess effect of fertilizers, nutrient omission trials, combinations of mineral and organic fertilizers, and how they can be managed under various cropping systems. Pot and field experiments are common (Johnston and Poulton 2018).

Animal Nutrition Experiments Mainly involve feeding trials designed to compare production results between two or more feeding schemes in livestock production. In most cases, the feeding experiments will aim to estimate digestibility and feed intake. This may include measuring weight, body length, food intake, palatability, and feeding preference (Meier et al. 2012).

Veterinary Clinical Experiments These experiments entail research that acquires data from animal patients in veterinary settings (Taylor and Meyer 2023). Clinical studies aim to enhance animal healthcare by identifying the most efficient therapies and practices for a particular condition or by advancing a basic understanding of the disease. There are two primary categories of clinical studies: clinical trials, also known as interventional studies and observational studies. In a clinical trial, veterinary patients may undergo specific interventions based on a research plan or protocol developed by the investigators, while in observational trials, the natural course of the disease or condition is observed and analyzed (see Chap. 12).

Agroforestry Experiments Agroforestry integrates agriculture and forestry to address environmental and socioeconomic objectives. The experiments conducted in agroforestry are diverse and are geared toward optimizing productivity. They include experiments that test and screen species, such as screening experiments to compare trees' adaptability and growth rates. Other agroforestry-based experiments include system management and examination of components interactions, seed testing, evaluation of germplasm and provenances, and the domestication and selection of superior plus trees for improvement. Methods for quantifying carbon stored in agroforestry trees is another crucial area of research (Atangana et al. 2014).

Randomized Controlled Evaluations These are field experiments used to evaluate the impact of development programmes or policy interventions. The population subjected to the program or policy intervention is randomly selected from the eligible population, and simultaneously, a control group is randomly chosen from the same eligible population. Measurements are done after the intervention period. One of the notable strengths of this approach lies in its robust ability to address questions of causality, offering evaluators and program implementers a clear understanding that the observed outcomes result from the intervention itself and not from any other factors. Suppose you want to assess the effectiveness of two training methods: A and B. You will need to choose two groups of farmers and assign each training method to them, including a control (not trained). You would have three treatments (Training method A, Training method B and control). The allocation of the treatments would be random. This is discussed in detail in Chap. 11.

Crop Breeding Experiments Crop breeding is a method of crop improvement. It involves changing crops' traits to bring on board desired characteristics intending to improve productivity. Most studied agricultural traits are those associated with biotic and abiotic stress tolerance, grain and biomass yield, taste and nutrition, and ease of processing. Usually, the process begins with developing inbred lines, which are then propagated for additional testing and eventual variety release. The predominant method can be categorized into three main groups: 1. Selection based on observed variation involves choosing plants with desirable traits found in the natural variation or within traditional varieties; 2. Controlled mating selects plants with desirable genes from parents through controlled breeding; 3. Monitored recombination selects specific genes or marker profiles using molecular tools to track variation within a genome (Hasan et al. 2021).

Biotechnology Experiments Biotechnology applies science and engineering to utilize living organisms or systems to create products and services (Ranjha et al. 2022). Experiments involve the development of genetically modified crops, medicines, biofuels and environmental tools, as well as using techniques like DNA sequencing, cloning and fermentation. A brief discussion of biotechnology is given in Chap. 17.

8.3 Fundamentals of Experimentation

8.3.1 Role of Experimentation

Two fundamental pillars of science are the meticulous design and execution of scientific research and the capability to replicate research findings, as highlighted in Chap. 1. Implementation rigor guarantees an impartial approach to experimental design, methodology, analysis, interpretation, and reporting of results (Hofseth 2018). Reproducibility validates original results. This aspect holds particular significance when disseminating results to end users.

Experiments play a central role in scientific study by providing a method to test cause-effect hypotheses. The success

of modern science owes much to the careful design and precise execution of experiments. As mentioned earlier, R.A Fisher, a geneticist and statistician, significantly contributed to the current theory and experimental methods. Experiments are usually associated with theory-testing research. Theories, instrumental in problem-solving, often outline the expected outcomes when a change is introduced. See the following:

- There is a problem with fruit flies in mangoes, but a theory tells us that a new mango variety is tolerant to fruit flies, so our prediction will be: If we use an improved mango variety, will there be less or no damage from fruit flies?
- Low milk yields on smallholder farms are a problem because of inadequate protein supplementation due to the high costs of dairy meal. However, a theory tells us that Calliandra fodder contains protein and can be used to substitute dairy meal. Will milk yields increase if we use Calliandra fodder to replace dairy meals?
- Farmers use low-quality cattle manure, resulting in low maize yields, but a theory exists that if farmers are trained to manage manure, the yield of maize will increase. If we train farmers on manure management, will maize yield increase?

To test a theory, you introduce the change and observe whether the predicted outcome occurs. In essence, we test the hypothesis (see Chaps. 2 and 20). This is the basis of experimentation, which entails implementing a specific action and methodically observing the resulting effects.

8.3.2 The Concept of Experimentation

Experimental research is characterized by manipulation, differentiating it from other research types. This manipulation involves deliberately altering one or more independent variables to observe their impact on the dependent variable. Researchers determine the nature of the treatment, who receives it (such as individuals), and the extent of its application (Kim et al. 2021). After administering the treatment for a specific duration, researchers observe or measure the groups subjected to different treatments (post-test) to determine whether the treatment has induced a discernible difference. The primary goal is to determine whether the treatment has had a meaningful impact. This differs from an observational study, which involves analyzing without changing existing conditions.

Consider an experimental research scenario in which two groups of farmers are subjected to different soil fertility treatments (treatment A and B). The researcher applies the treatments and conducts a post-test to measure crop yields. Suppose the average scores of groups A & B show significant differences and the researchers thoroughly investigate without finding any sensible alternative explanations (such as variations in weather conditions or pest infestations). In that case, they can reasonably conclude that the treatment affected crop yields.

There are circumstances in which experimentation is unrealistic and unethical. In these circumstances, alternative methods are used. It is not feasible to experiment with a prediction such as: if the average annual rainfall increases by 200 mm, maize production in country A will increase. We could test this prediction using simulation models describing the relationship between rainfall and maize production. However, these models will be based on theories tested by experiments.

Conversely, a correlational study might reveal a strong relationship between farmers' education levels (a socio-economic factor) and adopting hybrid maize varieties. However, the correlational study alone cannot demonstrate that improving farmers' education levels will necessarily lead to an enhancement in adoption of hybrid maize varieties. Only experimental research, with controlled manipulation of variables, can provide a more robust basis for making such a causal inference.

Social studies involving human beings and communities make predictions that cannot be tested by conventional experimentation. As an alternative, researchers often employ observational methods, observing and analyzing social phenomena without intentional interventions. Surveys serve as a valuable tool to collect precise data on attitudes, behaviors, and perceptions, enabling the examination of correlations and patterns in social dynamics (Chap. 12). While experimentation is limited in some social contexts, the use of observational approaches and surveys remains crucial for gaining precise insights into human behavior and societal trends (Babbie 2021).

8.3.3 The Experimental Process

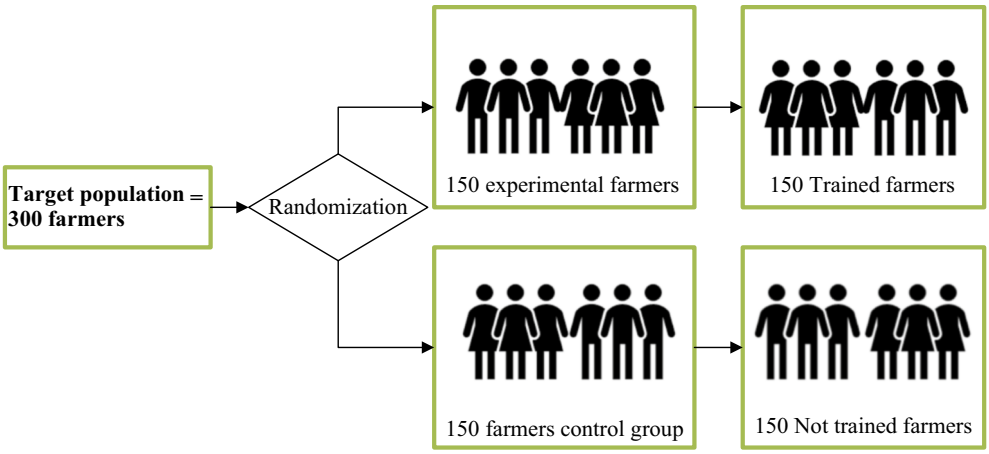
The procedures involved in an experimental study closely mirror those in other types of research types. They encompass selecting a topic and framing a problem, establishing objectives and hypotheses, crafting a research plan (selecting treatments, experimental units and measuring instruments), implementing the study, scrutinizing the data and ultimately formulating conclusions (Chap. 2). An experimental study is guided by at least one hypothesis that states an expected causal relationship between two variables. The researcher experiments to test the hypothesis, carefully selecting the groups and allocating the treatments. In addition, the researcher ensures that the extraneous variables are controlled to be able to attribute the observations (measurements) to the treatments.

In an experiment, the group that receives the new treatment is called the *experimental group*, and the group that receives a different treatment or is treated as usual is called the *control group*. The groups to receive the different treatments should be equated on all variables that may influence the performance of the dependent variable (Box 8.1).

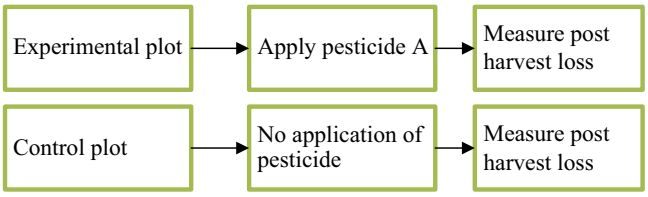
Using the example of farmers' training in Box 8.1, suppose at the end of the experimental period, trained farmers record an average post-harvest loss of 20%, while the other group (control) records an average of 25% post-harvest loss. There is a 5-point difference; is it meaningful, or is it just a chance difference due to measurement errors? Statistical

Box 8.1: Illustration of Basic Experimental Process

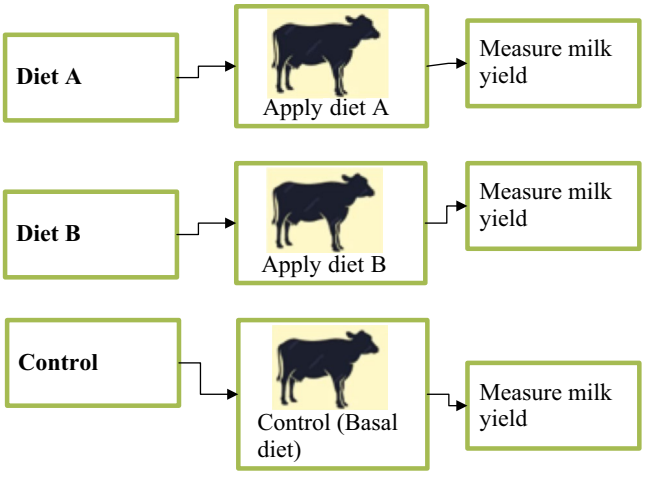
1. Research objective: To assess the effect of training farmers on the use of new, improved bags on post-harvest losses.



2. To assess the effect of pesticide A on yield of tomato.



3. To assess the effect of feeding dairy cows three different protein feed supplements (A control diet, Diet A and Diet B) on milk yields.



analysis allows the researcher to answer these questions with confidence. This is tackled in Chaps. 19 and 20.

8.3.4 The Concept of Variation and Experimental Error

As mentioned, experiments are powerful tools for researchers because they are objective. They carry the highest level of credibility among the research designs when properly designed and managed. Experiments allow us to study phenomena objectively. However, there is no perfect experiment that will explain the causal relationship. There will always be other things that can influence your dependent variable. We call these *extraneous variables or covariates* if measured (mentioned earlier in Chap. 2). In the real world, holding all the potential variables that could affect the dependent variable constant is impossible. We have *individual variation* (intrinsic) that occurs naturally. This intrinsic variation tends to obscure the effect of different interventions (treatments), making comparisons problematic. This can consequently lead to mistakes in interpreting the results of experiments, and subsequently erroneous judgments and recommendations for commercial purposes. Every researcher must thus understand this fundamental and universal concept of variation in scientific experimentation and their application (Mead et al. 2017).

We could think of the following variations that occur naturally:

- Experimental plots in the field vary greatly, and so do pots in a greenhouse experiment, even if we are using the same soil.
- Plants in an experimental plot vary, though in the same treatment application.
- Participants (human beings) in an experiment differ from each other in many different ways.
- Farmers are different from each other, first as individuals (human beings) and secondly in the way they manage their farms.
- The animals (cattle, goats, etc.) in an experiment vary in many ways, such as genetics and weight.
- A pair of twin heifers, being given the same diet in the same environment, will perform differently in growth as well as general characteristics.
- Similar seed lots of pumpkins planted on the same day, under the same environment; plants will show differences in growth and yields.
- Measurements taken on the same subject may vary naturally.

The causes of variation are numerous (Little and Hills 1978). For example, in the case of the growth of plants from the same seed lot, the list of factors that might be expected to affect the growth of the plant includes differences in seed weight, soil variation, depth of planting, competition from other plants, interaction with soil microbes, etc. At maturity, the weighing of different plants, the use of different machines and the subjective recording of the weights are also expected to result in variations. Similarly, a human or animal experiment will be expected to have numerous causes of variation.

In comparison, however, biological materials tend to exhibit a much greater amount of variation than physical science materials and processes (Brown 1993; Ilan 2020). This is mainly because of the genetic system and sensitivity of the living matter during its growth and development to the effects of the environment. Overall, no matter how many factors we attempt to control, we shall always find that the final weights of two plants will be different, even in a controlled environment.

All these variations and many more cause *random fluctuations* that have nothing to do with your independent variable. All these variations are causes of error in experiments, referred to as *random variation* or *experimental error*. In other words, experimental error is a result of variation caused in our dependent variable by things that we are not able to explain because we cannot measure them directly. On the other hand, we directly measure variation in the dependent variable due to treatments. Experimental error is *unexplainable*. How, then, do we handle this error in our experiments? This is the basis of understanding the fundamentals and principles of experimentation and application of statistical science.

The purpose of applying statistical procedures in experiments is to separate the unexplained part of the variation from the variation due to treatments (i.e. due to the independent variable) (Little and Hills 1978). At the beginning of this Chapter, it was referred to as *partitioning the data variation*. Statistical analysis procedures allow researchers to mathematically identify and isolate this effect that may have been caused by other variables, so that the true treatment effects (true value) are more clearly detectable. The statistical analysis also allows the researcher to conclude whether the observed differences are real and then assigns a probability level that your conclusion is correct. The probability level is incorporated as a measure of accuracy (Chap. 20 on inferential analysis).

In summary, we see that there are two main sources of variation in an experiment:

1. *Variation due to treatments*: Variation in the dependent variable resulting from the application of distinct treatments for which you can account in the statistical analysis; also called variation due to treatments or explained variation.
2. *Variation due to non-treatment factors*: Variation in the dependent variable that cannot be accounted for; also

called un-explained variation. It includes all other sources of variation in the dependent variable that does not directly result from the experimental treatments. It is called "Noise" because it obscures the effect of the treatment (discussed later in this Chapter).

The concepts discussed in the rest of this Chapter and Chap. 9 are important to help a researcher minimize the unexplained variation and maximize the explainable variation.

8.4 General Principles of Experimentation

The three main principles of experimentation, namely randomization, replication and local control (reducing noise), are known as the 3 Rs.

8.4.1 Principle of Randomization

What is Randomization?

Randomization is the random assignment of treatments or conditions of the study to the experimental group without any prior plan, that is, by chance (Hirst et al. 2014). The law of probability is applied. Every experimental unit has an equal chance of being allocated to any of the experimental units. An experimental unit is the entity that carries the treatment.

What Is the Purpose?

Random assignment avoids experimenter bias. It randomly assigns treatments to experimental units and helps distribute unknown extraneous variables evenly. This implies that it prevents systematic and subjective bias from the experimenter. For example, an experimenter may tend to favor an innovation or variety. A researcher working with a new bean variety carries out tests compared to other older varieties, but assigns the new variety to fertile fields. Another essential purpose of randomization is that it forms a basis for valid statistical tests. Independent observations are one of the preconditions for subjecting data to statistical tests e.g. ANOVA and t-tests. The lack of random assignment makes comparisons biased no matter how precise the measurements are.

How Do You Randomize?

Random allocation may be done by drawing lots, such as a ball from a container or numbers from a table of random numbers. Methods of randomization are covered in later sections dealing with individual forms of experimental designs.

8.4.2 Principle of Replication

What Is replication?

Replication means repetition. In this case, the study's treatments or conditions are repeated in experimental units (Demétrio et al. 2013). The end result of replication is several units of each treatment.

What Is the Purpose?

- Enables a valid estimate of experimental error, allowing statistical inferences, such as performing tests of significance. It is impossible to estimate experimental error without replication. This is important for *estimating precision*, that is, the ability to quantify uncertainty. Replicates are needed to distinguish "noise from signal" (unexplained variation from variation due to treatments) or estimate the precision of treatment differences. There is no way of estimating error from the un-replicated experiment. In experiments, we want to determine the effect of treatments (control variables) on the dependent variable, hence detecting this signal.
- *Decrease experimental error*: Reduced error leads to increased precision—a measure of the variability of the experimental error. Calculating an average over more values will increase precision and reliability since the calculated value will be closer to the true mean.
- *Replication* enhances the validity of results by repeating the comparison under various conditions, thereby expanding the range of applicability or generalization. This practice enables a more accurate estimation of the mean effect of a treatment.

Determining the appropriate number of replications in any experiment depends on several factors, including the homogeneity of the experimental material, the number of treatments, and the desired level of precision. Increasing the number of replications reduces experimental error, indicating that a higher number leads to greater precision in the experiment. As a general guideline, it is recommended that the number of replications in a design should offer at least 10–15° of freedom for computing the experimental error variance (Patel et al. 2004).

8.4.3 Principle of Local Control or Reducing Noise

Local control is an additional technique to the randomization and replication to control the experimental error further and increase the experiment's precision (Hirst et al. 2014). It per-

tains to the strategic implementation of balancing, blocking and grouping the experimental units to account for the variability in response due to the variability caused by inherent differences in experimental units, and the individual items that receive the treatment. Blocking involves physically *grouping treatments* in an experiment. Balancing in experimental design involves assigning treatments to experimental units in a manner that results in a well-distributed or balanced arrangement of treatments. This practice ensures that each treatment is equally represented and distributed across the experimental units, contributing to a more equitable and unbiased assessment of the treatment effects.

What Is the Purpose?
The primary objective is to enhance efficiency by minimizing experimental error. Blocking measures and isolates block variation. By doing so, the goal is to extract this *block-level variation* from the statistical comparison of treatment means. This process enables a more accurate assessment of treatment effects by reducing the interference caused by extraneous sources of variability (experimental error). This ultimately improves the precision and reliability of experimental results.

How Do you Carry Out Local Control?
If you can anticipate sources of variation, you can group the treatments to decrease variation within each group and reduce some variation from the statistical analysis. Blocking should be done in the field so that the "Blocks" are *homogenous*. Blocking is usually done against slopes, soil fertility gradients, or other forms of variation in the field. The treatments are "*nested*" within a homogenous environment. Researchers sometimes carry out uniformity trials to help determine the extent of soil homogeneity.

Relationship Between Blocks and Replications
Experimental designs typically use blocks consisting of one plot for each experimental treatment. For example, a block containing four treatments would include four plots, each receiving a different treatment. When a block has one plot per treatment, it represents one replication of each treatment. Therefore, blocks are often called "replications" informally. It is important to note that "replication" and "blocking" are two distinct concepts. Replication involves repeating the entire experiment to improve statistical reliability, whereas blocking aims to *minimize variability within specific groups* or blocks of experimental units.

8.4.4 Noise Versus Signal

Experimental error is also called "noise" because it makes it difficult to hear the signal (Marchant et al. 2019; Berry et al. 2022). It is caused by unexplained or uncontrolled variation. In a crop experiment, plots treated alike may give different responses due to numerous factors that, if not correctly controlled, tend to increase the experimental error. These include:

1. *Human error*: Human errors in measurements and observations can be caused by various factors, including variations in the skills, interpretations and subjective judgments of individual operators collecting data. Additionally, inaccuracies may arise due to imprecision or limitations of the measuring instruments used in the experiment. Addressing and minimizing these human errors is crucial in ensuring the accuracy and reliability of the results. Therefore, it is essential to provide *proper training, standardize procedures* and carefully select and *calibrate* measuring devices to achieve this goal.
2. *Variability among plots*: Inherent variability among plots, including distinctions in soil characteristics such as drainage, elevation and compaction, constitutes a significant source of variation in an experiment. Researchers often conduct preliminary assessments, such as *uniformity trials*, to address this challenge before initiating the main experiment. By characterizing and understanding the inherent variability within the experimental site, researchers can implement strategies to control or account for these factors during the main experiment.
3. *Lack of uniformity in the physical conduct of the experiment*. For instance, an unequal distribution of fertilizer doses across plots can introduce variability, which may affect the experiment's reliability. This non-uniformity can arise from factors such as carelessness in applying treatments. Similarly, variations in animal characteristics in animal experiments can contribute to inconsistencies in the experimental conditions. Following rigorous *standardization protocols* and paying meticulous attention to detail is essential to avoid such issues. Ensuring uniformity in the application of treatments and maintaining consistent conditions across experimental units are critical steps in reducing the impact of these potential sources of variability on the experiment's outcomes.
4. *Contamination from neighboring plots*: Interference among treatments in adjacent plots causes variation. For example, suppose irrigation is done in one plot. In that case, it might cause fertilizer or rhizobia (in the case of inoculation experiments) treatments to percolate into the neighboring plots, resulting in unintended cross-contamination. To address this issue, it is crucial to carefully consider the spacing, buffer zones and appropriate experimental design (Discussed in Chap. 9).
5. *Non-uniform behavior of all other extraneous factors*, known or unknown, that have not been/could not be con-

trolled. For example, weather changes can affect the experiment's reliability by introducing difficult-to-manage variability. To address this issue, researchers often employ statistical techniques, randomization, and replication techniques to account for the impact of uncontrollable factors. While it may not be possible to control every external variable, these strategies enhance the experiment's robustness by statistically accounting for the influence of uncontrolled factors on the observed outcomes.

Measures to prevent experimental error should be a top priority to ensure that the observed effects can be accurately attributed to the intended treatments without external interference.

Box 8.2: The Concept of "Noise" and Its Control in an Experiment

"Noise" in experiments refers to the unexplained variation in observed outcomes that cannot be directly attributed to the treatments or factors being studied. This interference can arise from differences among experimental units, uncontrolled environmental conditions, measurement errors or other unforeseen factors. Noise obscures the "signal" just like background noise in a classroom corridor can make it difficult to hear what the teacher is saying. When noise is present in an experiment, we cannot hear the signal. The noise introduces random fluctuations or disturbances in the data, making it difficult to distinguish between what is genuinely caused by the *treatments* and what is merely the result of *unforeseen factors*. This interference acts like background "noise" in the data, *masking or clouding* the true signal generated by the treatment.

For example, in experiments that test the effect of different fertilizers on crop yield, the signal is the actual change in crop yield that can be attributed to the fertilizers. However, noise from differences in soil fertility, can distort the signal by introducing variability in the observed yields that is not directly related to the fertilizers. As a result, the true impact of fertilizers on crop yield can become obscured by this unwanted variation, and the real effects of the fertilizer can be hidden or clouded by all these other factors. It will be difficult to detect significant differences in an experiment, with most variation being in the residuals (error) in ANOVA.

Reducing noise in an experiment is crucial for revealing the clear signal from the treatments. The goal is to enhance the *signal-to-noise ratio*, to hear the signal. We employ the strategies such as randomization, replication, control of experimental conditions, etc., standardizing processes. When we do this, we can confidently say, "Yes, the fertilizer is making a positive difference," without being confused by all the other stuff going on in the background.

8.5 Elements of Designing Effective Agricultural Experiments

In addition to the application of the three principles of experimentation (the 3Rs), designing an effective agricultural experiment requires you to consider the following 15 elements:

1. Objectives and hypotheses.
2. Experimental design.
3. Treatments.
4. Experimental units.
5. Number of replications.
6. Control treatment/controlling variables.
7. Experimental site selection.
8. Uniformity trials.
9. Management of the experiment.
10. Response variables.
11. Precise measurements.
12. Experimental protocol.
13. Recording data.
14. Data analysis and interpretation.
15. Conclusion and recommendation.

1. *Objectives and Hypotheses*

   Always recall that "objectives drive the study" and therefore, all aspects of the design depend on objectives; in addition to being SMART (See Chap. 3), they must therefore be:

   (a) *Complete*: indicate both the independent (treatments) and dependent variables (parameters to be measured); if not complete, it is impossible to design the experiment. An objective stating "the objective is to compare treatments" or an objective is to "evaluate the performance of varieties" is incomplete.

   (b) *Relevant*: should be relevant to solving the problem of your study. They should link clearly to the research problem and the results expected to provide the solution.

   (c) *Capable of being met by an experiment*: The nature of the objective and variables should be capable of being met by an experiment.

2. *Design of Experiment*

The experiment should be designed and conducted so that there will be a high probability of measuring differences between treatments with acceptable precision (Campbell and Stanley 1963). Ensure you apply the basic principles of experimentation. Many designs are available to choose from depending on the number of variables and are discussed in the next Chapter.

3. *Treatments*

The choice of treatments will depend on objectives. The following should be considered when choosing treatment:

- *Quantitative levels*: Some experiments require varying quantities that could have many different levels, e.g. sowing dates, amount of fertilizer to be applied, e.g. comparing the effect of fertilizer N levels on yields. Choosing the level to use depends on the objective (emanated from the research gap) and what you already know about the response. For example, in an experiment to determine the effect of different fertilizer levels on the yield of maize, where the issue is to determine optimal levels for N for farmers, the levels chosen will depend on the levels the farmers are using, the recommended rate and then the new levels you wish to compare (Box 8.3).
- *Comparison and contrasts*: Experiments involve comparisons; ensure all the treatments needed to make all comparisons implied by the objectives are included.
- *Factorial treatment structure*: Many experimental objectives involve investigating interactions by examining several treatment factors. Ensure all relevant combinations of treatments are included.

Box 8.3: Determining Fertilizer Levels in Experiments

Suppose you are conducting an experiment to identify the optimum level that maximizes crop yields; It is important to check recommended fertilizer application rates and methods. You also need to check local agricultural practices and expert recommendations from extension services or experienced farmers in the region to ensure that their rates and practices are factored into the rates in the experiment. It is recommended to test the levels of fertilizer covering a range from low to high. For instance, you can consider levels such as 0 kg/ha (control, no fertilizer), 50 kg/ha, 100 kg/ha, 150 kg/ha and 200 kg/ha. Ensure incremental increases that are uniform, for example, you may increase fertilizer by 25 kg/ha or 50 kg/ha.

4. *Experimental Units*

The experimental unit is the fundamental entity to which a specific treatment is applied, whether a *person, object, plant, tree, animal, plot or pot* (Kim et al. 2021). This essential element of experimental design represents a subset chosen from a broader population of interest. For instance, in agricultural research, plots within a field are commonly selected as experimental units. These plots represent the larger farming area, and their treatments provide insights into how the interventions may affect the entire population. Table 8.1 shows different types of experimental units.

5. *Number of Replications*

As we have seen, all experiments must be replicated. In addition to the other three reasons given earlier for replication, replication in agriculture has an additional advantage of *insurance*; more replications provide insurance against things going wrong, e.g. crops being destroyed, farmers

Table 8.1 Various combinations of independent variables showing possible experimental units

Independent variable (factor)	Treatments	Experimental unit	Dependent variable
Treated vs. *treated (effect of treatment IV on DV)*	*Intervention*	*Treated and non-treated plots*	*Results being measured*
Fertilizer (Effect of fertilizer levels on crop yields)	Levels of fertilizer	Pots	Crop yield
Animal diets Effect of protein diets on milk yields	Types of diets	Animals	Milk yields
Training (effect of training on uptake of technology)	Trained or untrained	Farmers	Uptake of technology (numbers and extent)
Pesticides Effect of pesticide on yields of crops	Types of pesticides	Plots	Yields, economics
Training on manure used (effect of training on knowledge levels of manure)	Trained and untrained	Farmers	Knowledge levels on the use of manure
Vaccine (Effect of vaccine on disease incidences of goats	Vaccine types	Goats	Number of diseased vs non-diseased

dropping out of the trial, animals dying, etc. The number of replications will be determined by the site factors (land availability, resources, existing variability, etc.). The more replications, the more land is required. A greenhouse experiment may even have 10 replications. Field experiments mainly have three to six replications. Trials with farmers need more replications due to high farm variability (Njuho 1998; Patel et al. 2004).

There should be enough replicates to satisfy all the reasons for replication. Ten is considered a reasonable minimum. Very many replicates have no advantage. The number required for insurance depends on the risk. A long-term trial in a risky environment may be worth ensuring by adding more replications (Watson et al. 2024). This is the case with on-farm trials discussed in Chap. 10. In on-farm experiments, we deliberately use more replicates.

The number of replications is a common question that students ask. Determining the number of replications is often a decision involving trade-offs between statistical rigor and practical constraints. Most authors (Berndtson 1991; Patel et al. 2004; Aaron and Hays 2004; Hall 2007; Gent et al. 2018) determine the number of replications based on the following:

- Use of power analysis.
- Previous studies and pilot studies.
- Resource constraints.
- Expected biological variability and desired precision.

Fig. 8.2 A control plot next to a treated plot for comparison; in a soil fertility experiment in Tharaka-Nithi county, Kenya

In most crop, soil, fodder and forage experiments, researchers have primarily utilized 3, 4 or 5 replications as these numbers strike a balance between statistical rigor and practical considerations. The choice is primarily based on a literature review of similar studies, but researchers also consider the number of treatments, and resource constraints, including land, budget, and available labor. The availability of land and the possibility of introducing more variability in many treatments are always major considerations in determining the number of treatments and replication to manage variability in various experimental conditions effectively.

6. *Control of Environment and Control Treatment*

What is control, and why is it necessary? We need to ensure two types of controls in experiments: controlled environment and controlled treatment as discussed below:

1. *Controlled variables*: this requires researchers to keep conditions that are not being investigated constant. This encompasses elements like temperature, humidity, soil conditions and other environmental variables that may vary across experimental units. This allows researchers to isolate the effects of the specific treatments they are studying. For instance, in a fertilizer trial, the same cultivars should be used, and all non-treatment factors must be kept constant.
2. *Control treatment*: This is a treatment incorporated into the experiment and used to establish a *baseline or reference point* against which the effects of other treatments can be compared (Fig. 8.2). The control treatments usually involve maintaining the status quo or applying a standard practice, i.e. the independent variable is kept at a normal level. For example, in an experiment assessing the effect of a diet fortified with vitamin A on the growth of animals, the control recipients could have the normal diet used by farmers. The selection of control treatment is a critical decision that highly depends on the objective of the study. You should always consider suitable controls, which are not necessarily treatment without any intervention (see Box 8.4).

Box 8.4: Selecting Control Treatment

Control treatment may not mean "zero input" (Patel et al. 2004). Farmers' practice could be the most appropriate. At times, you could include two controls, which may include a farmer practice and an absolute control (no inputs). See the following examples.

Fertilizer experiment
Research objective: Assess the effectiveness of various fertilizer formulations on rice yields under smallholder farms. Suitable control could have the following:

- Plot with no input (zero inputs)
- Farmer practice: plot with farmers conventional fertilizer use

Crop protection experiment
Research objective: Assess the efficacy of different pesticides in controlling early blight in tomatoes. Suitable control could have the following:

- A group (plot) with no pesticide application
- Placebo; apply water or an inert material
- Farmer practice: Conventional pesticides used by farmers

Animal nutrition experiment
Research objective: Assess the effect of different crude protein diets on milk yields in dairy cows. Suitable control could have the following:

- A group of cows on a well-balanced diet
- A group of cows on a farmer's conventional diet (farmers' practice)

7. *Experimental Site*

The site for the experiment will be determined by the research objectives. It has to represent the problem area on a large scale, e.g. in the same agroecological zone. Representativeness on a small scale considers such things as appropriate soil types and previous management. Accessibility, security and adequacy of land are important parameters to consider. The objectives will also dictate the number of sites for the experiment. For instance, if the experiment requires comparing conditions, such as soil type and rainfall, more than one site will be needed.

In site selection, aim to keep sources of variation at a minimum. Avoid fields with poor drainage, ditches, slopes, etc., and prioritize having a homogenous field. The field's history is important, as it helps us know which areas to avoid. Soil fertility gradients are common, especially in smallholder farms. In some cases, researchers may grow catch crops (e.g. non-fertilized maize) to remove soil nutrients, especially nitrogen.

8. *Uniformity Trials*

Uniformity trials are conducted before setting the experiment to assess a particular factor's uniformity and variability across a plot or field. These trials are crucial for controlling spatial variability, one of the main factors that determines the efficiency of experiments (Casler 2015; Jones et al. 2021). They are common in crop breeding and agronomy experiments. Uniformity trials provide valuable information that guides subsequent experimental design and statistical analysis. This helps to minimize bias and random errors caused by uneven treatment application.

In a soil fertility study, a uniformity trial would be a critical component enabling researchers to determine soil fertility variation of a targeted experimental field. A common practice is to plant maize in a field without applying fertilizer. As the crop is growing, assessments are made to determine areas of low and high soil fertility based on the maize growth. The field is divided at harvest into small units (plots and yields recorded). From these yields, a researcher can draw a "fertility contour map" which gives a graphical picture of the variation in soil fertility (Jones et al. 2021). Eventually, this information is used to subdivide the field into relatively homogenous sub-groups to control experimental errors that are critical in designing the experiment. The homogenous groups often form the blocks.

9. *Management of Experiments*

In a field experiment, management means preparing the land, sowing, weeding and all other agronomic practices needed to raise crops, deworming animals, and spraying animals. It is usually not considered as part of the design but can have a large impact on the data, yet they have a large impact on the success of the experiment. Always aim for uniform management of the animals, crops or people for all variables that are not part of the treatments. This will play a key role in *preventing confounding treatments* with management practices. For example, in the management of agricultural experiments, all management practices should be applied following recommended practices except for the treatment variable, e.g. if testing the performance of 6-series maize varieties in each region, the spacing, fertilizer application or input application should be uniform and according to the recommended practices.

10. *Precise Measurements*

Taking precise measurements is key to obtaining accurate and reliable data. Precise measurements reduce measurement errors and enhance the validity of the experimental results. Errors in experiments are mainly from the measurements and positional conditions of the experimental unit or from randomization of the experimental units (Hartung et al. 2019). During planning (in the protocol) clearly indicate the data that will be collected and how each will be collected. Have a data collection schedule that specifies the timing and frequency of data collection, e.g. data may be collected daily (milk yields), weekly or at specific growth stages.

11. *Response Variables*

Several variables describe the experimental units in relation to some of their characteristics and we usually distinguish (i) response variables, (ii) factor variables, and (iii) covariates. The response variables, also called outcome variables, describe the response of units to the experimental treatments (e.g. the yield, the weight, the height, and so on), while the factor variables describe the experimental stimulus allocated to each unit (e.g. fertilizer rate, genotype, herbicide applied, etc.) (Coe et al. 2003, 2014). In some cases, we also record other covariates, also called accessory variables, which measure potential confounding effects. For example, if we intend to study the yield of maize under different fertilizer levels, the soil parameters can act as a confounder. Other covariates, depending on the objective of the experiment, may include pest and disease incidences. Growth parameters, phenological stages data, e.g. flowering and fruiting, may also be taken.

We also collect other types of data depending on the nature of the experiment. *Characterization data* is used to contextualize the site and determine the recommendation domain. These include the location of the experiment, rainfall, soil characteristics, dates of planting, etc.

In any experiment, data collection typically occurs in three main phases. First, before the experiment begins with the aim of assess the appropriateness of the site for research. This involves evaluating soil quality, climate conditions and other environmental variables to ensure they align with the research objectives. This phase also establishes a baseline for comparison and provides context for interpreting subsequent measurements. The second phase involves collecting measurements during the experiment itself. This includes recording various parameters such as labor use, dry matter, degree of damage, and disease incidences throughout the experiment. Finally, at the conclusion of the experiment, measurements are taken to assess outcomes such as crop yields, milk yields and the impact on farm and household incomes. Collecting data at each of these phases is essential for understanding the progression of the experiment, evaluating its outcomes and drawing meaningful conclusions.

12. *Experimental Protocol*

An experimental protocol is an important tool that guides research. It is a written description of everything that will be done in the experiment, starting with the objectives and analysis, interpretation and use of data (Casler 2015; Giraldo et al. 2018; Sichinga-Ligowe et al. 2024). Protocols act as a roadmap for research. Within research organizations, each study has its protocol, typically stored in a database. Often, it is a requirement that management approves the protocol before implementation. This approval process serves as a guide and minimizes the likelihood of errors during implementation. Furthermore, written protocols ensure continuity, particularly in cases of staff turnover. By adhering to protocols, researchers maintain consistency and transparency in their work, facilitating the sharing of findings with fellow scientists. Ultimately, protocols are crucial in guiding research, ensuring data reliability and enhancing collaboration within the scientific community.

In academic proposals, protocols should be included to capture detailed information about the study (Sichinga-Ligowe et al. 2024). This may be included in the protocols in the appendix of the research proposal, as university guidelines often limit the space available for detailed protocols within the main document. The protocol ensures that essential information is accessible without overwhelming the main proposal. Besides helping you plan your research, the protocol is a valuable tool for supervisors to evaluate the research critically. This ensures correct data collection and implementation of the research work. Moreover, in collaborative projects, it may be necessary for each study to have a protocol included in the project file. If in such a project, you must prepare one for your academic research.

A sound experimental protocol must fulfill the following purposes:

- It allows you to include details; it may not be possible to write everything in the research proposal within the word limit.
- Ensures that the experiment is conducted consistently, thus reducing variability during data collection.
- Allows other researchers to replicate the study.
- Allow other researchers to take over if the principal investigator is not in a position to continue, e.g. in case of transfer or natural attrition.
- Record keeping serving as a permanent record of methods and procedures used.

Key Components of a Crop Experiment Protocol

- Introduction: indicate the research objective and hypothesis to be tested, and include background information on the crop, soil and agricultural systems.
- Materials and methods; list all the materials and equipment required for the experiment. Include specifications and quantities, e.g. types of fertilizer and quantities to be applied, crop varieties and source of seeds.
- Experimental design; indicate the experimental design, treatments and number of replications. Explain how randomization will be done and also have a field plan and plot layout plan. Show the size of the plots and the size of the paths and border areas.
- Methods: outline step-by-step procedures to be followed during the experiment. Include information on soil preparation, application of inputs, planting and crop protection. Explain in detail all the agronomic procedures to be followed.
- Data collection methods: explain details of data collection methods. Indicate the data to be collected and how it will be collected. Indicate the step-by-step process of how each data will be collected and recorded. Be specific and include time frames for the various activities. Include how harvesting will be done with the.
- Data analysis: specify the statistical methods of data analysis techniques that will be used to analyze the data. Include statistical tests, software, models, data standards, etc.
- Safety precautions: Describe safety measures and precautions to ensure the well-being of the researchers, workers, data collection technicians and farmers. Include guidelines for handling chemicals and any hazardous materials.
- Timelines: include a timeline for when each phase of the experiment will occur.
- Appendices: include any additional information, documents, data collection forms, consent forms and standard operating procedures.

A researcher, whether a graduate student or a principal investigator, must ensure the protocol is sufficiently detailed, kept up to date and securely archived so that information about the activity can be found in the future and that a biometrician is involved (Patel et al. 2004).

13. *Recording Data*

In all kinds of experiments, data is collected through direct observation and measurements. During data collection, record all the data in a manner that ensures accuracy and consistency using data sheets or digital tools. In addition, regularly calibrating the measuring instruments and equipment, scales, rulers, soil testing devices, soil pH meters, etc., ensures that the technicians collecting the data are well-trained on proper techniques and procedures for making measurements. Use well-designed data sheets that are complete, clear and easy to use.

At the end of the experiment, data should be organized and stored securely ensuring that it is well-documented and easily accessible for analysis and future reference.

During data analysis, data should be checked for accuracy and consistency to identify anomalies and outliers that may need verification and correction.

14. *Data Analysis and Interpretation*

At the end of the study, data analysis is conducted to interpret the results and draw meaningful conclusions. The analysis often requires the use of statistical techniques to make informed recommendations or determine the appropriate course of action (Gomez and Gomez 1984). The method of statistical analysis employed depends on both the objective of the study and the experimental design used. As stated at the beginning of this chapter, statistical analysis in experiments primarily involves *partitioning the sources of variation* in the data to isolate treatment effects from random error. Chapter. 9 shows how the sources of variation are partitioned depending on the design. The researcher must conduct a thorough and appropriate statistical analysis to ensure that the findings are both valid and reliable. For detailed procedures and examples, refer to Chaps. 20 and 21 on experimental data analysis, which guide conducting and interpreting statistical tests for various experimental designs.

15. *Conclusions and Recommendations*

At the reporting stage, researchers organize their findings systematically, known as the experiment's conclusions. The conclusions summarize the discoveries, patterns and relationships while indicating whether they align with the initial hypotheses. Clear and concise conclusions are vital for conveying the significance of the experiment's outcomes and acknowledging any limitations. Ultimately, the conclusions provide a brief overview of the experiment's impact on the research question.

Recommendations guide future research, decision-making and real-world applications. Clear and well-supported recommendations enhance the experiment's usefulness, guiding subsequent studies and facilitating the implementation of research findings. Researchers may also highlight areas where further research is needed (Covered in Part 6 on Reporting).

Recommended Further Reading

1. Campbell DT, Stanley JC. 1963. Experimental and Quasi-Experimental Designs for Research.
2. The Green Book: A guide to effective graduate research in African Agriculture by Patel et al. (2004).
3. Gomez KA, Gomez AA. 1984. Statistical procedures for agricultural research. Statistical procedures for agricultural research with emphasis on rice. (September): 7–8.
4. Little TM., Hills F Jackson. 1978. Agricultural experimentation: design and analysis https://www.wiley.com/en-us/Agricultural+Experimentation%3A+Design+and+Analysis-p-9780471023524
5. Coe, R.D., et al. (2002). Designing Experiments and Analyzing Data. Lecture Notes. World Agroforestry Centre (ICRAF) publication.

Key Points in This Chapter

Brief History of Experiments

R.A. Fisher, a biostatistician, developed experimental design and analysis techniques while at Rothamsted Experimental Station. These methods contributed to the field of experimentation and have been widely adopted in research.

Classes of Experiments

There are two broad classes of agricultural experiments: (1) laboratory experiments conducted in a controlled, artificial setting to control possible extraneous variables and (2) field experiments carried out in the natural environment. Experiments can either be carried out on an on-station site where the researcher solely manages the experiment or in the on-farm environment on the farmer's fields, and the farmer being involved.

Common Types of Field Experiments

Common types of field experiments include the following:

- *Agronomy experiments*: Aim is to develop management practices that optimize the production of the crops.
- *Varietal screening experiments*: Test the performance of different varieties to compare their performance across some characteristics.
- *Forage production experiments*: Aim to test different forage varieties, management practices and environmental conditions to determine the best strategies for maximizing yields and quality of forages.
- *Intercropping experiments*: designed to help understand interactions between species so that we can design systems that enhance resource utilization.
- *Crop rotation experiments*: investigate how different rotation patterns can be manipulated to enhance agricultural productivity.
- *Soil fertility experiments*: Evaluate the effect of soil management practices on the availability of soil nutrients and agricultural productivity.
- *Animal nutrition experiments*: Feeding trials to compare the effects of different diets on livestock performance.
- *Veterinary clinical experiments*; Aim to advance animal health care by identifying the most effective therapies and practices for a given condition.
- *Agroforestry experiments*: develop integrated systems of agriculture and forestry components that are productive, profitable, healthy and sustainable.
- *Randomized controlled evaluations*: Randomized experiments aimed to evaluate the impact of development programs or policy interventions on specific outcomes.
- *Crop breeding experiments* involve changing crop traits to incorporate desired characteristics to enhance yield, resistance, and adaptability.
- Biotechnology experiments: apply advanced biological techniques to increase agricultural productivity.

General Principles of Experimentation

There are three main principles of experimentation, namely: randomization, replication, and local control (reducing noise). They are also known as the 3 Rs. *Randomization* refers to *the* random assignment of treatments or conditions of the study. *Replication* means repeating treatments or study conditions. *Local control* is a technique that involves blocking and grouping experimental units into homogeneous entities.

Noise Versus Signal

Experimental error is also called "noise" because it makes it difficult to hear the signal (detect response). In a crop experiment, the experimental error is likely to be from (1) human while making observations and measurements, (2) inherent variability among plots, e.g. the soil differences, such as drainage, elevation and compaction, (3) lack of uniformity in the physical conduct of the experiment, (4) interference among treatments in neighboring plots, and (5) non-uniform behavior of all other extraneous factors.

Elements of Designing Effective Agricultural Experiments

In addition to the application of the three principles of experimentation (the 3Rs), designing an effective agricultural experiment requires you to consider the following key elements:

1. Objectives and hypotheses.
2. Experimental design.
3. Treatments.
4. Experimental units.
5. Number of replications.
6. Control (Treatment and controlling the variables).
7. Uniformity trials.
8. Site selection.
9. Management of the experiment: Always aim for uniform management of the animals, crops, or people for all variables that are not part of the treatments. This will play a key role in preventing confounding treatments with management practices.
10. Experimental protocol.
11. Taking precise measurements is key to obtaining accurate and reliable data. Precise measurements reduce measurement errors and enhance the validity of the experimental results.
12. Response variables: distinguish (i) response variables, (ii) factor variables, and (iii) accessory variables.
13. Recording data.
14. Data analysis and interpretation.
15. Conclusions and recommendations.

Review Questions

1. Who was R.A. Fisher, and what was his significant contribution to the field of experimentation?
2. Explain the "noise versus signal" analogy in experimental error. What is the "signal" you are trying to detect in a crop experiment, and what constitutes the "noise" (experimental error)?
3. Identify and explain the three main principles of experimentation (the 3 Rs).
4. Why is uniform management of an experiment an important principle?
5. Explain how to determine the number of replications.
6. Why is developing a detailed experimental protocol essential before commencing any agricultural experiment?

Chapter Exercise

1. Identify and describe three distinct sources of experimental error ("noise") in a crop experiment. For each, suggest a practical measure a researcher could take during the experiment's management to minimize that specific error.
2. Choose a recent peer-reviewed journal article that reports the results of an agricultural experiment in your area of specialization. After reading the article, identify how the authors applied the three core principles of experimental design: randomization, replication, and local control (blocking). Were these principles clearly stated and appropriately implemented? Give a personal critique.

Answers to Review Questions

- Q1: Reference 8.1.1
  - Laid the foundation principles for experimentation
  - Introduced the concept of 3R
  - Developed ANOVA
  - Provided a systematic and objective approach to scientific inquiry
- Q2: Reference 8.3.4
- Q2: Randomization, Replication and educe Noise - Local Control (3R). Reference 8.4.1, 8.4.2, 8.4.3
  - Randomization involves random assignment of treatments to experimental units. This helps to evenly distribute the unknown extraneous variables.
  - Replication: Repeating each treatment. It increases precision and reliability, allows estimation of experimental error, and enhances generalization.
  - Local; control: groups experimental units in more homogenous blocks or entities; this reduces experimental error, increases precision, and improves efficiency.
- Q4: Reference 8.5 (9)
- Q5: Reference 8.5 (5)

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